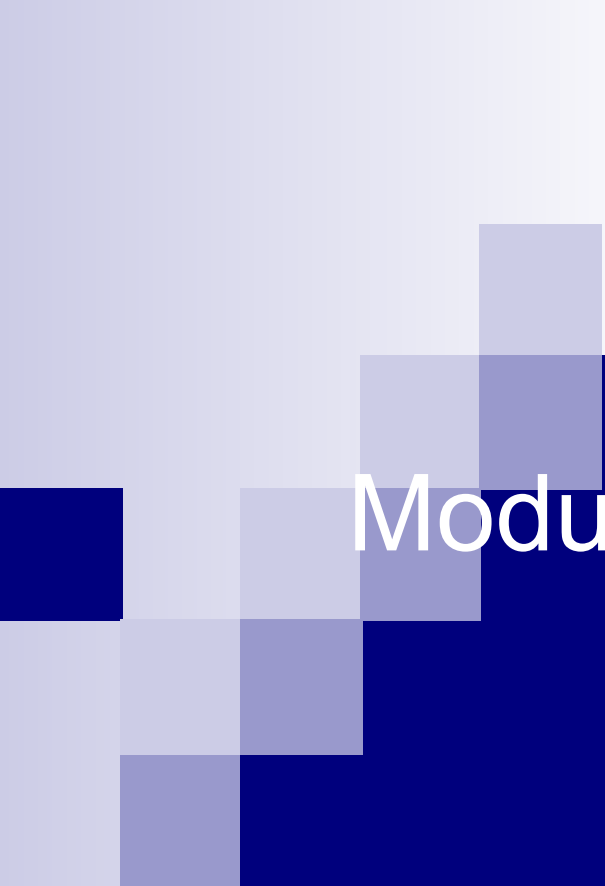


Cloud Computing (21CS643)



Module -1



Introduction

- Cloud computing is a technology that uses the internet and central remote servers to maintain data and applications.
- It allows consumers to use applications without installation and access their personal files at any computer with internet access.

Life before cloud computing

- Traditional business applications have always been very complicated and expensive. The amount and variety of hardware and software required to run them are daunting. You need whole team of experts to install, configure, test, run, secure and update them.

Life after cloud computing

- With cloud computing, you eliminate those headaches because you're not managing hardware and software—that's the responsibility of an experienced vendor like salesforce.com. The shared infrastructure means it works like a utility: You only pay for what you need, upgrades are automatic, and scaling up or down is easy.

Cloud computing

- Uses Internet technologies to offer scalable and elastic services. The term “elastic computing” refers to the ability of *dynamically acquiring computing resources* and supporting a variable workload.
- The resources used for these services can be metered and the *users can be charged only for the resources they used*.
- The maintenance and security are ensured by service providers.

Cloud computing (cont'd)

- Lower costs for the cloud service provider are passed to the cloud users.
- Data is stored:
 - closer to the site where it is used.
 - in a device and in a location-independent manner.
- The data storage strategy can increase reliability, as well as security, and can lower communication costs.

Types of clouds

- Public Cloud - the infrastructure is made available to the general public or a large industry group and is owned by the organization selling cloud services.
- Private Cloud – the infrastructure is operated solely for an organization.
- Community Cloud - the infrastructure is shared by several organizations and supports a community that has shared concerns.
- Hybrid Cloud - composition of two or more clouds (public, private, or community) as unique entities but bound by standardized technology that enables data and application portability.

The “good” about cloud computing

- Resources, such as CPU cycles, storage, network bandwidth, are shared.
- When multiple applications share a system, their peak demands for resources are not synchronized thus, *multiplexing leads to a higher resource utilization*.
- Resources can be aggregated to support data-intensive applications.
- Data sharing facilitates collaborative activities. Many applications require multiple types of analysis of shared data sets and multiple decisions carried out by groups scattered around the globe.

“good” about cloud computing

- Eliminates the initial investment costs for a private computing infrastructure and the maintenance and operation costs.
- Cost reduction: concentration of resources creates the opportunity to pay as you go for computing.
- Elasticity: the ability to accommodate workloads with very large peak-to-average ratios.
- User convenience: virtualization allows users to operate in familiar environments rather than in idiosyncratic ones.

Why cloud computing could be successful when other paradigms have failed?

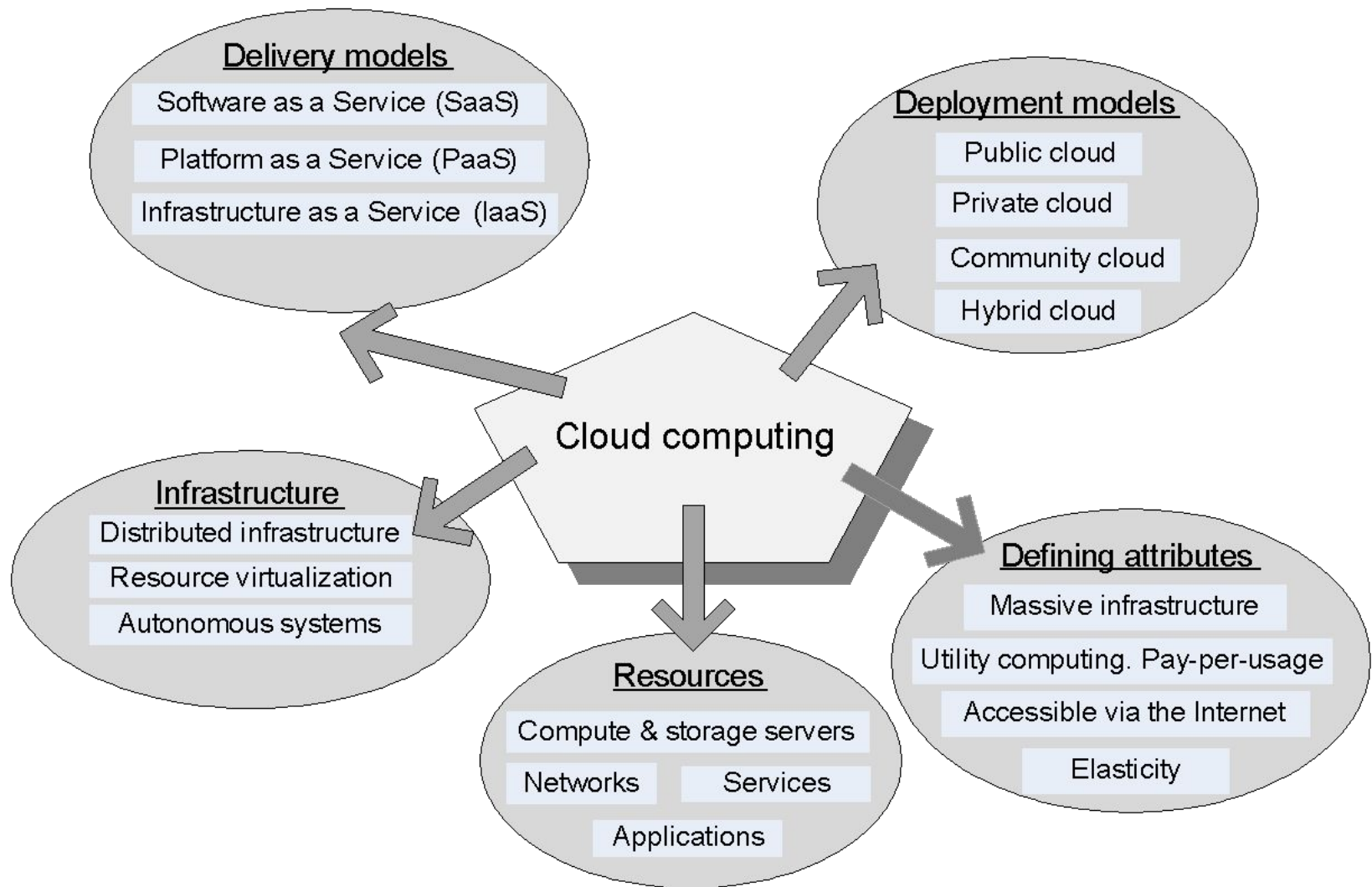
- It is in a better position to exploit recent advances in software, networking, storage, and processor technologies promoted by the same companies who provide cloud services.
- It is focused on enterprise computing; its adoption by industrial organizations, financial institutions, government, and so on could have a huge impact on the economy.
- A cloud consists of a homogeneous set of hardware and software resources.
- The resources are in a single administrative domain (AD). Security, resource management, fault-tolerance, and quality of service are less challenging than in a heterogeneous environment with resources in multiple ADs.

Challenges for cloud computing

- Availability of service; what happens when the service provider cannot deliver?
- Diversity of services, data organization, user interfaces available at different service providers limit user mobility; once a customer is hooked to one provider it is hard to move to another.
Standardization efforts at NIST!
- Data confidentiality and auditability, a serious problem.

More challenges

- Performance unpredictability, one of the consequences of resource sharing.
 - How to use resource virtualization and performance isolation for QoS guarantees?
 - How to support elasticity, the ability to scale up and down quickly?
- Resource management; are self-organization and self-management the solution?
- Security and confidentiality; major concern.
- Addressing these challenges provides good research opportunities!!





Cloud delivery models

- Software as a Service (SaaS)
- Platform as a Service (PaaS)
- Infrastructure as a Service (IaaS)

Software-as-a-Service (SaaS)

- SAAS sometimes referred to as “Software on demand”, is software that is deployed over the internet.
- Applications are supplied by the service provider.
- The user does not manage or control the underlying cloud infrastructure or individual application capabilities.
- Examples: Gmail, Google search engine.

Advantages

- Accessible from anywhere with an internet connection.
- No local server installation.
- Pay per use or subscription-based payment methods
- System maintenance (backup, updates, security, etc) often included in service

Cloud delivery models

Example: Pizza Delivery 🍕

- **① Infrastructure as a Service (IaaS) – "You Make the Pizza at Home"**
 - You **rent the infrastructure** (oven, gas, electricity) but cook yourself.
 - **Example: Amazon Web Services (AWS EC2)** – You rent virtual machines but manage everything else.



② Platform as a Service (PaaS) – "You Buy a Frozen Pizza"

- The base is ready; you just **bake and add toppings**.
- **Example: Google App Engine** – You build an app but hosting and database are managed for you.

③ Software as a Service (SaaS) – "You Order a Ready-Made Pizza"

- Everything is done; you just **eat the pizza**.
- **Example: Google Docs** – You use it online without installing anything.

Platform-as-a-Service (PaaS)

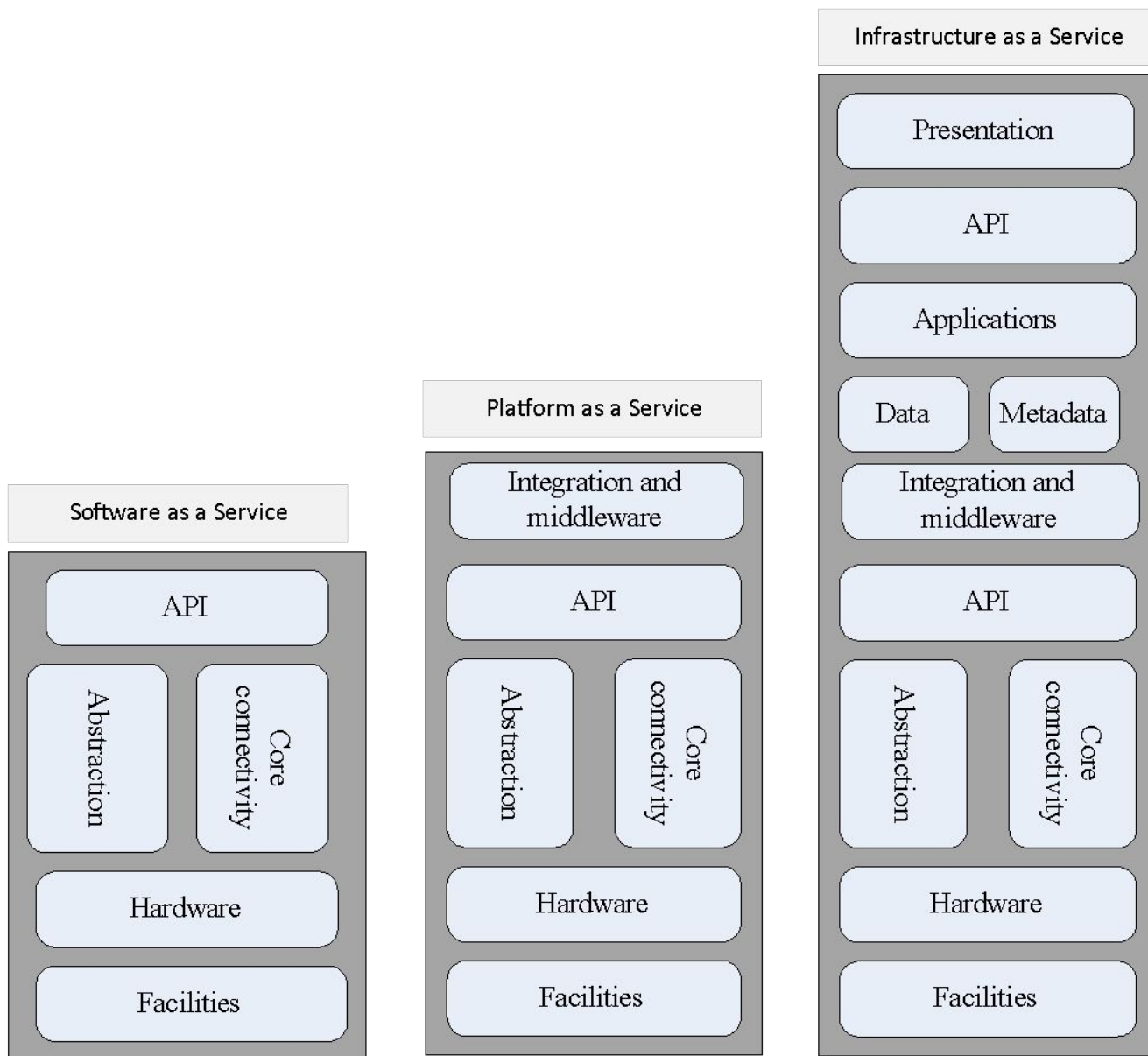
- Used by developers
- It provides a platform and environment to allow developers to build applications and services over the internet
- Offers development and deployment tools required to develop applications
- The user:
 - Has control over the deployed applications and, possibly, application hosting environment configurations.
 - Does not manage or control the underlying cloud infrastructure including network, servers, operating systems, or storage.

Platform-as-a-Service (PaaS)

- Allows a cloud user to deploy consumer-created or acquired applications using programming languages and tools supported by the service provider.
- Not particularly useful when:
 - The application must be portable.
 - Proprietary programming languages are used.
 - The hardware and software must be customized to improve the performance of the application.

Infrastructure-as-a-Service (IaaS)

- The user is able to deploy and run arbitrary software, which can include operating systems and applications.
- The user does not manage or control the underlying cloud infrastructure but has control over operating systems, storage, deployed applications, and possibly limited control of some networking components, e.g., host firewalls.
- Services offered by this delivery model include: server hosting, Web servers, storage, computing hardware, operating systems, virtual instances, load balancing, Internet access, and bandwidth provisioning.



Cloud activities

- Service management and provisioning including:
 - Virtualization.
 - Service provisioning.
 - Call center.
 - Operations management.
 - Systems management.
 - QoS management.
 - Billing and accounting, asset management.
 - SLA management.
 - Technical support and backups.

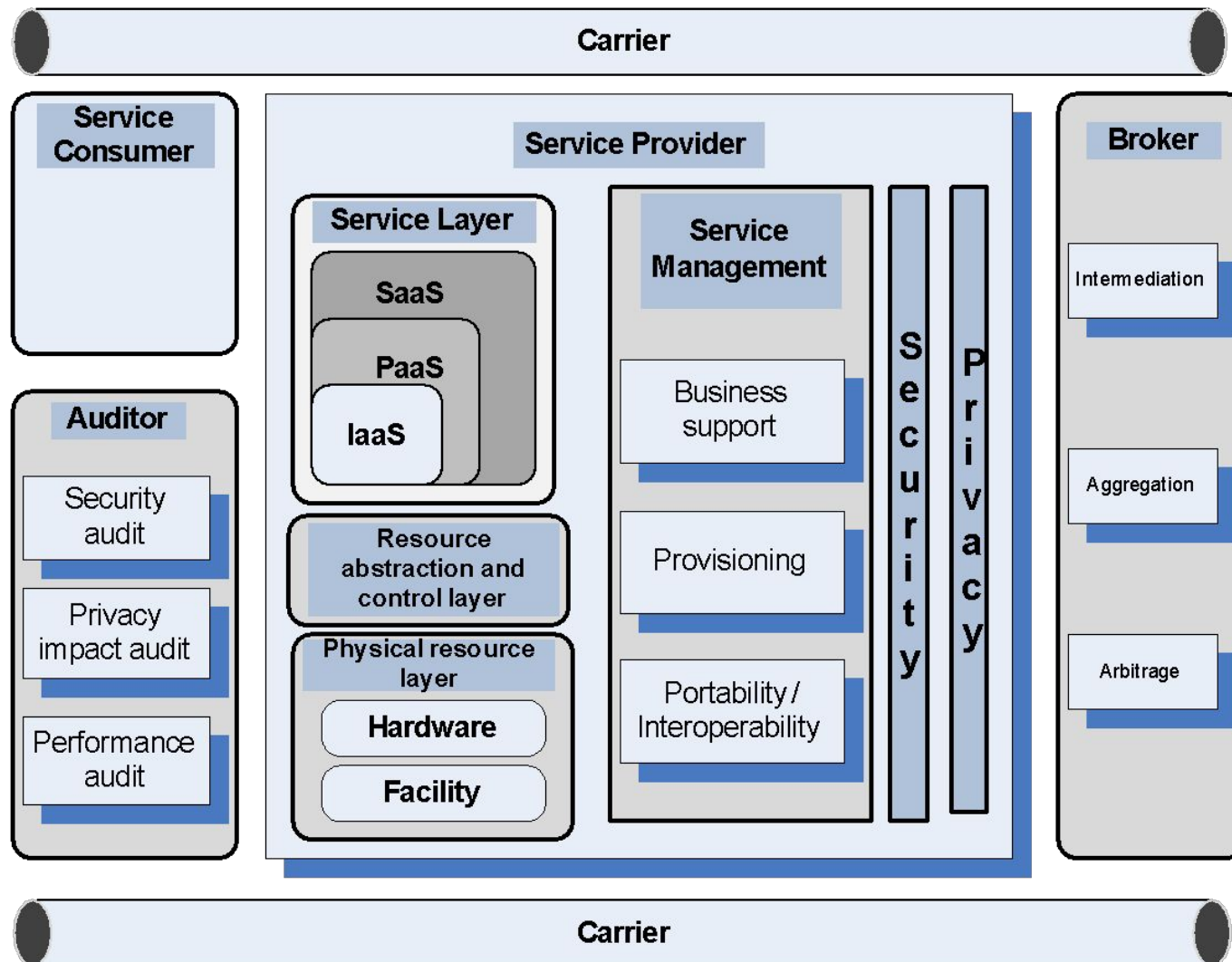
Cloud activities (cont'd)

- Security management including:
 - ID and authentication.
 - Certification and accreditation.
 - Intrusion prevention.
 - Intrusion detection.
 - Virus protection.
 - Cryptography.
 - Physical security, incident response.
 - Access control, audit and trails, and firewalls.

Cloud activities (cont'd)

- Customer services such as:
 - Customer assistance and on-line help.
 - Subscriptions.
 - Business intelligence.
 - Reporting.
 - Customer preferences.
 - Personalization.
- Integration services including:
 - Data management.
 - Development.

NIST cloud reference model



NIST Cloud Reference Model –

- The National Institute of Standards and Technology (NIST) created a model to explain how cloud computing works. Think of it like an online food delivery system 🍔📦, where different people and systems work together to get your food (or cloud service) delivered.

- ① Cloud Consumer (The Customer – You!)
- The person or company using the cloud.
- Example: A student saving assignments on Google Drive, or a business using Microsoft 365.
- 💡 In a food delivery app: You place an order for food (just like you use a cloud service).

-  Cloud Provider (The Restaurant – Service Owner)
- The company that owns and manages the cloud infrastructure.
- Example: Amazon Web Services (AWS), Google Cloud, Microsoft Azure.
-  In food delivery: The restaurant prepares and delivers your meal (just like the provider runs the cloud service).

- ③ Cloud Carrier (The Delivery Partner – Internet Service)
- The network or internet that connects users to the cloud.
- Example: Your WiFi or mobile data that lets you access cloud apps.
- 💡 In food delivery: The delivery person (Zomato, Swiggy, Uber Eats) brings your food from the restaurant to your home (just like the internet connects you to cloud services).

- **④ Cloud Broker (The Best Deal Finder – Consultant)**
- Helps businesses pick the best cloud service by comparing prices, features, and quality.
- Example: A company that helps a hospital decide between AWS or Google Cloud for storing patient data.
-  In food delivery: A price comparison app that helps you find the best restaurant with discounts and offers.

-  Cloud Auditor (The Quality Checker – Security Inspector)
 - Ensures security, performance, and compliance with rules.
 - Example: An independent company checking if Google Cloud keeps user data safe.
 -  In food delivery: A food safety officer ensuring the restaurant follows hygiene standards.

Ethical issues

- Paradigm shift with implications on computing ethics:
 - The control is relinquished to third party services.
 - The data is stored on multiple sites administered by several organizations.
 - Multiple services interoperate across the network.
- Implications
 - Unauthorized access.
 - Data corruption.
 - Infrastructure failure, and service unavailability.

The complex structure of cloud services can make it difficult to determine who is responsible in case something undesirable happens

De-perimeterisation

- protect an organisation's systems and data on multiple levels, by using a mixture of encryption, secure computer protocols, secure computer systems and data-level authentication”
- The complex structure of cloud services can make it difficult to determine who is responsible in case something undesirable happens.
- Identity fraud and theft are made possible by the unauthorized access to personal data in circulation and by new forms of dissemination through social networks and they could also pose a danger to cloud computing.

Privacy issues

- Cloud service providers have already collected petabytes of sensitive personal information stored in data centers around the world. The acceptance of cloud computing therefore will be determined by privacy issues addressed by these companies and the countries where the data centers are located.
- Privacy is affected by cultural differences; some cultures favor privacy, others emphasize community. This leads to an ambivalent attitude towards privacy in the Internet which is a global system.

Cloud vulnerabilities

- Clouds are affected by malicious attacks and failures of the infrastructure, e.g., power failures.
- Such events can affect the Internet domain name servers and prevent access to a cloud or can directly affect the clouds:
 - in 2004 an attack at Akamai caused a domain name outage and a major blackout that affected Google, Yahoo, and other sites.
(Akamai, a content delivery network (CDN) provider, experienced a domain name outage due to a targeted attack. This outage affected several major websites, including Google and Yahoo, which relied on Akamai's services to deliver content efficiently to users.)

Cloud vulnerabilities

- in 2009, Google was the target of a denial-of-service attack which took down Google News and Gmail for several days;
- in 2012 lightning caused a prolonged down time at Amazon.

The lightning strike caused power outages and equipment failures, disrupting services for AWS customers.

Amazon Web Services (AWS)

- AWS □ IaaS cloud computing services launched in 2006.
- Businesses in 200 countries used AWS in 2012.
- The infrastructure consists of compute and storage servers interconnected by high-speed networks and supports a set of services.
- An application developer:
 - Installs applications on a platform of his/her choice.
 - Manages resources allocated by Amazon.

AWS regions and availability zones

- Amazon offers cloud services through a network of data centers on several continents.
- In each *region* there are several availability zones interconnected by high-speed networks.
- An *availability zone* is a data center consisting of a large number of servers.

Region	Location	Availability zones	Cost
US West	Oregon	us-west-2a/2b/2c	Low
US West	North California	us-west-1a/1b/1c	High
US East	North Virginia	us-east-1a/2a/3a/4a	Low
Europe	Ireland	eu-west-1a/1b/1c	Medium
South America	Sao Paulo, Brazil	sa-east-1a/1b	Very high
Asia Pacific	Tokyo, Japan	ap-northeast-1a/1b	High
Asia Pacific	Singapore	ap-southeast-1a/1b	Medium

- Regions do not share resources and communicate through the Internet.

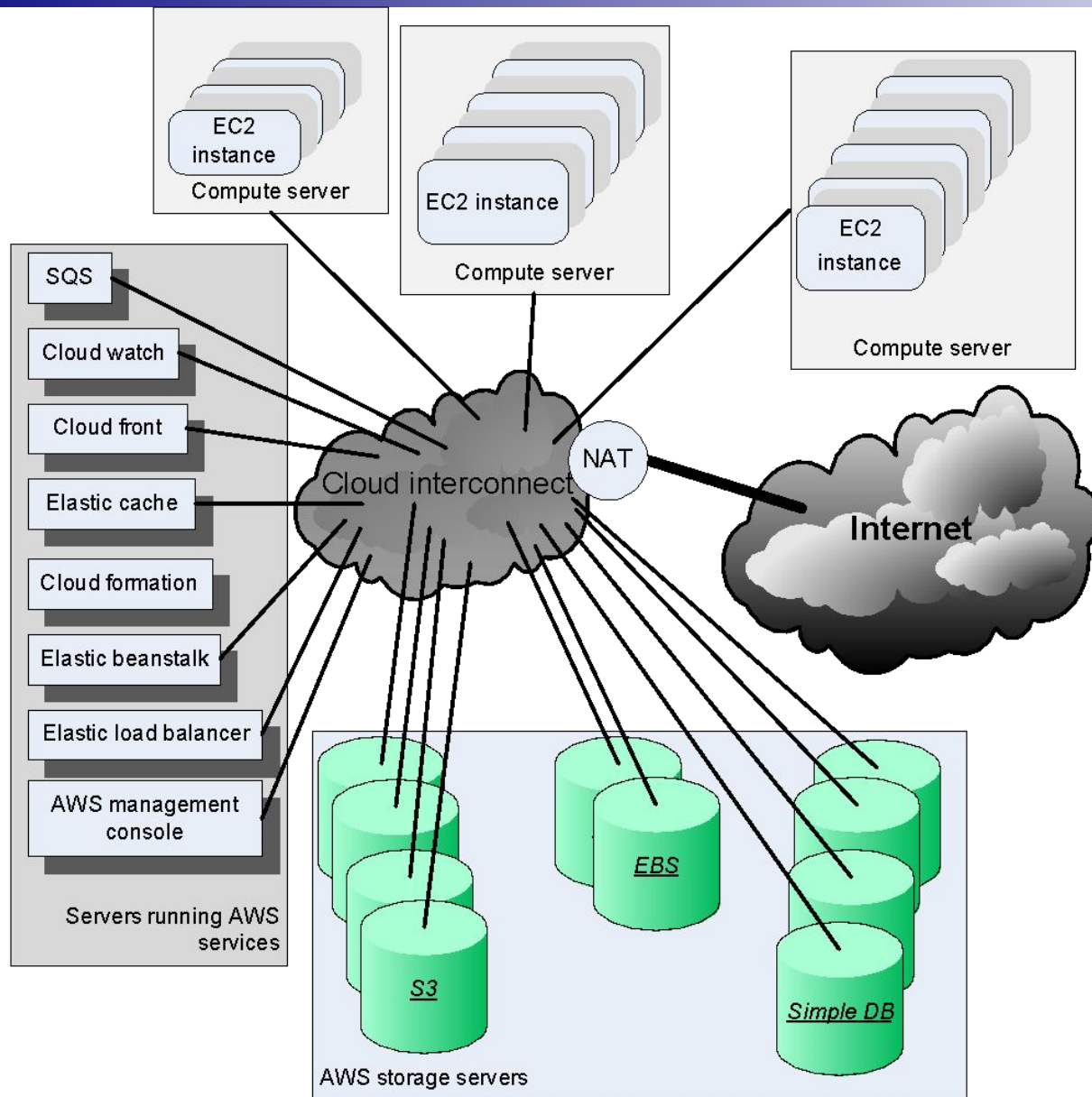


AWS instances

- An instance is a virtual server with a well specified set of resources including CPU cycles, main memory, secondary storage, communication and I/O bandwidth.
- The user chooses:
 - The region and the availability zone where this virtual server should be placed.
 - An instance type from a limited menu of instance types.
- When launched, an instance is provided with a DNS name; this name maps to a
 - *private IP address* □ for internal communication within the internal EC2 communication network.
 - *public IP address* □ for communication outside the internal Amazon network, e.g., for communication with the user that launched the instance.

AWS instances (cont'd)

- Network Address Translation (NAT) maps external IP addresses to internal ones.
- The public IP address is assigned for the lifetime of an instance.



Steps to run an application

- Retrieve the user input from the front-end.
- Retrieve the disk image of a VM (Virtual Machine) from a repository.
- Locate a system and requests the VMM (Virtual Machine Monitor) running on that system to setup a VM.
- Invoke the Dynamic Host Configuration Protocol (DHCP) and the IP bridging software to set up MAC and IP addresses for the VM.

User interactions with AWS

- The AWS Management Console. The easiest way to access all services, but not all options may be available.
- AWS SDK libraries and toolkits are provided for several programming languages including Java, PHP, C#, and Objective-C.
- Raw REST requests.

Examples of Amazon Web Services

- *AWS Management Console* - allows users to access the services offered by AWS .
- *Elastic Cloud Computing (EC2)* - allows a user to launch a variety of operating systems.
- *Simple Queuing Service (SQS)* - allows multiple *EC2* instances to communicate with one another.
- *Simple Storage Service (S3), Simple DB, and Elastic Block Store (EBS)* - storage services.
- *Cloud Watch* - supports performance monitoring.
- *Auto Scaling* - supports elastic resource management.
- *Virtual Private Cloud* - allows direct migration of parallel applications(provides a bridge between existing IT infrastructure of an organization and the AWS cloud).

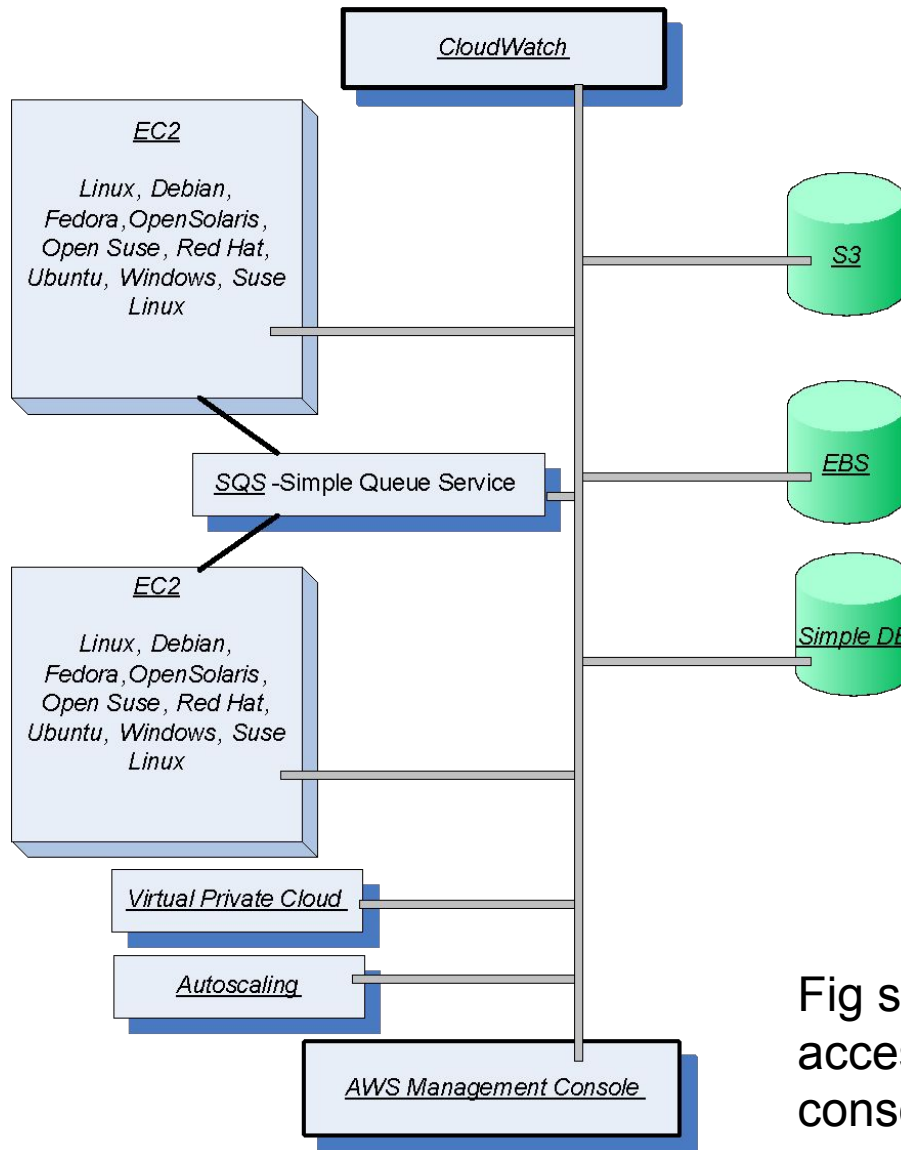


Fig shows the AWS services accessible via Management console

EC2 – Elastic Cloud Computing

- *EC2* - web service for launching instances of an application under several operating systems, such as:
 - Several Linux distributions.
 - Microsoft Windows Server 2003 and 2008.
 - OpenSolaris.
 - FreeBSD.
 - NetBSD.
- A user can
 - Load an *EC2* instance with a custom application environment.
 - Manage network's access permissions.

EC2 (cont'd)

- Import virtual machine (VM) images from the user environment to an instance through *VM import*.
- EC2 instances boot from an AMI (Amazon Machine Image) digitally signed and stored in S3.
- Users can access:
 - Images provided by Amazon.
 - Customize an image and store it in S3.

Instance types

- Standard instances: micro (StdM), small (StdS), large (StdL), extra large (StdXL); small is the default.
- High memory instances: high-memory extra large (HmXL), high-memory double extra large (Hm2XL), and high-memory quadruple extra large (Hm4XL).
- High CPU instances: high-CPU extra large (HcpuXL).
- Cluster computing: cluster computing quadruple extra large (Cl4XL).

Instance name	API name	Platform (32/64-bit)	Memory (GB)	Max EC2 compute units	I-memory (GB)	I/O (M/H)
StdM		32 and 64	0.633	1 VC; 2 CUs		
StdS	m1.small	32	1.7	1 VC; 1 CU	160	M
StdL	m1.large	64	7.5	2 VCs; 2 × 2 CUs	85	H
StdXL	m1.xlarge	64	15	4 VCs; 4 × 2 CUs	1,690	H
HmXL	m2.xlarge	64	17.1	2 VCs; 2 × 3.25 CUs	420	M
Hm2XL	m2.2xlarge	64	34.2	4 VCs; 4 × 3.25 CUs	850	H
Hm4XL	m2.4xlarge	64	68.4	8 VCs; 8 × 3.25 CUs	1,690	H
HcpuXL	c1.xlarge	64	7	8 VCs; 8 × 2.5 CUs	1,690	H
Cl4XL	cc1.4xlarge	64	18	33.5 CUs	1,690	H

Instance cost

- A main attraction of the Amazon cloud computing is the low cost.

Instance	Linux/Unix	Windows
StdM	0.007	0.013
StdS	0.03	0.048
StdL	0.124	0.208
StdXL	0.249	0.381
HmXL	0.175	0.231
Hm2XL	0.4	0.575
Hm4XL	0.799	1.1
HcpuXL	0.246	0.516
C14XL	0.544	N/A

S3 – Simple Storage System

- Service designed to store large objects; an application can handle an unlimited number of objects ranging in size from 1 byte to 5 TB.
- An object is stored in a bucket and retrieved via a unique, developer-assigned key; a bucket can be stored in a Region selected by the user.
- Supports a minimal set of functions: write, read, and delete; it does not support primitives to copy, to rename, or to move an object from one bucket to another.
- The object names are global.
- S3 maintains for each object: the name, modification time, an access control list, and up to 4 KB of user-defined metadata.

S3 (cont'd)

- Authentication mechanisms ensure that data is kept secure.
- Objects can be made public, and rights can be granted to other users.
- S3 computes the MD5 of every object written and returns it in a field called ETag.
- A user is expected to compute the MD5 of an object stored or written and compare this with the ETag; if the two values do not match, then the object was corrupted during transmission or storage.

Elastic Block Store (EBS)

- Provides persistent block level storage volumes for use with *EC2* instances; suitable for database applications, file systems, and applications using raw data devices.
- A volume appears to an application as a raw, unformatted and reliable physical disk; the range 1 GB -1 TB.
- An *EC2* instance may mount multiple volumes, but a volume cannot be shared among multiple instances.
- EBS supports the creation of snapshots of the volumes attached to an instance and then uses them to restart the instance.
- The volumes are grouped together in Availability Zones and are automatically replicated in each zone.

SimpleDB

- Non-relational data store. Supports store and query functions traditionally provided only by relational databases.
- Supports high performance Web applications; users can store and query data items via Web services requests.
- It manages automatically:
 - The infrastructure provisioning.
 - Hardware and software maintenance.
 - Replication and indexing of data items.

SQS - Simple Queue Service

- Hosted message queues are accessed through standard SOAP and Query interfaces.
- Supports automated workflows - *EC2* instances can coordinate by sending and receiving SQS messages.
- Applications using SQS can run independently and asynchronously, and do not need to be developed with the same technologies.
- A received message is “locked” during processing; if processing fails, the lock expires and the message is available again.
- Queue sharing can be restricted by IP address and time-of-day.

CloudWatch

- Monitoring infrastructure used by application developers, users, and system administrators to collect and track metrics important for optimizing the performance of applications and for increasing the efficiency of resource utilization.
- Without installing any software a user can monitor either seven or eight pre-selected metrics and then view graphs and statistics for these metrics.
- When launching an Amazon Machine Image (AMI) the user can start the CloudWatch and specify the type of monitoring:
 - Basic Monitoring - free of charge; collects data at five-minute intervals for up to seven metrics.
 - Detailed Monitoring - subject to charge; collects data at one minute interval.

AWS services introduced in 2012

- *Route 53* - low-latency DNS service used to manage user's DNS public records.
- *Elastic MapReduce (EMR)* - supports processing of large amounts of data using a hosted Hadoop running on *EC2*.
- *Simple Workflow Service (SWF)* - supports workflow management; allows scheduling, management of dependencies, and coordination of multiple *EC2* instances.
- *ElastiCache* - enables web applications to retrieve data from a managed in-memory caching system rather than a much slower disk-based database.
- *DynamoDB* - scalable and low-latency fully managed NoSQL database service.

AWS services introduced in 2012 (cont'd)

- *CloudFront* - web service for content delivery.
- *Elastic Load Balancer* - automatically distributes the incoming requests across multiple instances of the application.
- *Elastic Beanstalk* - handles automatically deployment, capacity provisioning, load balancing, auto-scaling, and application monitoring functions.
- *CloudFormation* - allows the creation of a stack describing the infrastructure for an application.

Elastic Beanstalk

- Handles automatically the deployment, capacity provisioning, load balancing, auto-scaling, and monitoring functions.
- Interacts with other services including *EC2*, *S3*, *SNS*, Elastic Load Balance and AutoScaling.
- The management functions provided by the service are:
 - Deploy a new application version (or rollback to a previous version).
 - Access to the results reported by CloudWatch monitoring service.
 - Email notifications when application status changes or application servers are added or removed.
 - Access to server log files without needing to login to the application servers.
- The service is available using: a Java platform, the PHP server-side description language, or the .NET framework.

Cloud computing : the Google perspective

SaaS services offered by Google

- *Gmail* - hosts Emails on Google servers and provides a web interface to access the Email.
- *Google docs* - a web-based software for building text documents, spreadsheets and presentations.
- *Google Calendar* - a browser-based scheduler; supports multiple user calendars, calendar sharing, event search, display of daily/weekly/monthly views, and so on.
- *Google Groups* - allows users to host discussion forums to create messages online or via Email.
- *Picasa* - a tool to upload, share, and edit images.
- *Google Maps* - web mapping service; offers street maps, a route planner, and an urban business locator for numerous countries around the world

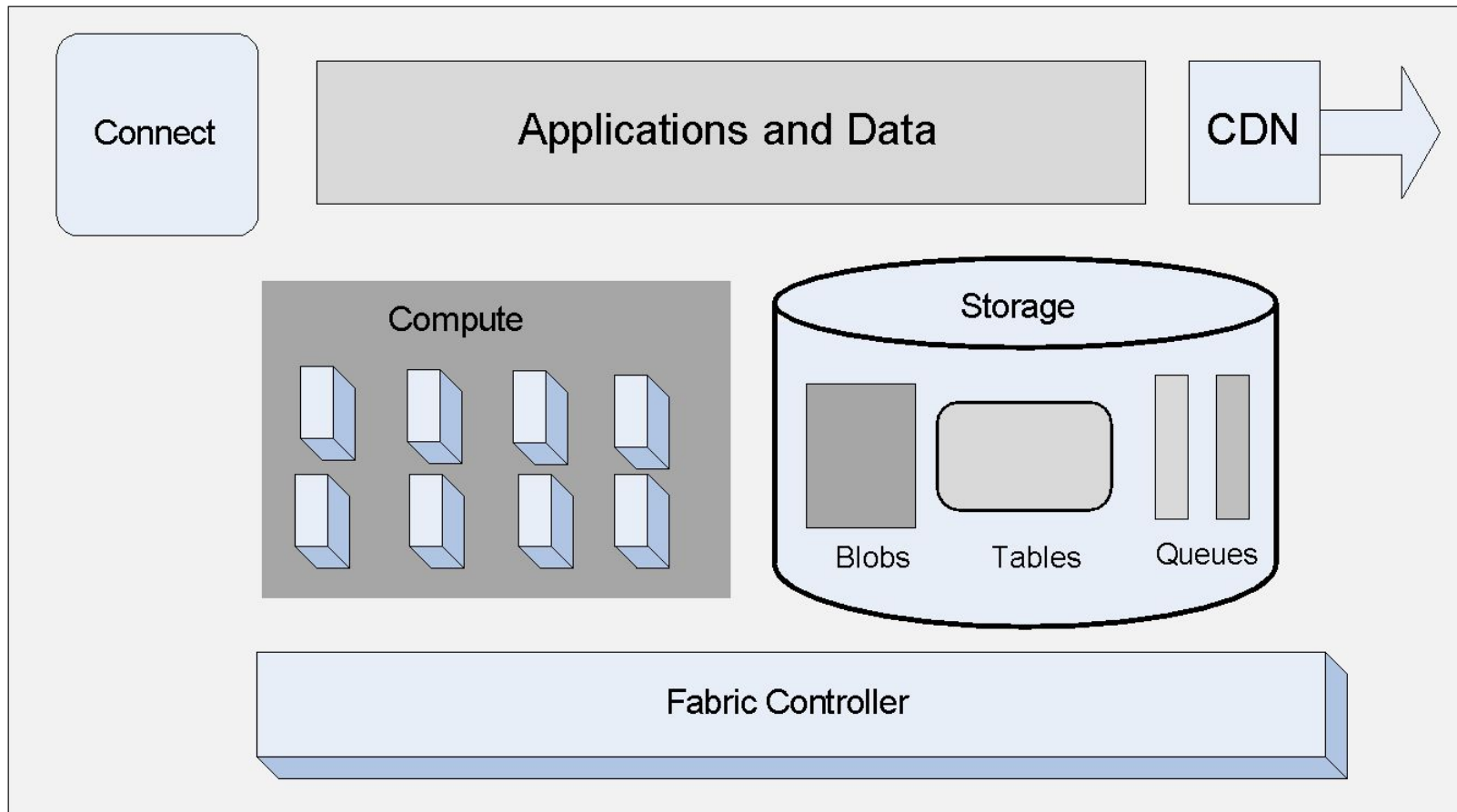
PaaS services offered by Google

- *AppEngine* - a developer platform hosted on the cloud.
 - Initially supported Python, Java was added later.
 - The database for code development can be accessed with GQL (Google Query Language) with a SQL-like syntax.
- *Google Co-op* - allows users to create customized search engines based on a set of facets/categories.
- *Google Drive* - an online service for data storage.

PaaS and SaaS services from Microsoft

- Designed by Microsoft in 2010, Microsoft Azure is one of the widely used cloud computing platforms. Azure provides a wide variety of services such as cloud storage, compute services, network services, cognitive services, databases, analytics, and IoT
- *Windows Azure* - an operating system; has 3 components:
 - Compute - provides a computation environment.
 - Storage - for scalable storage.
 - Fabric Controller - deploys, manages, and monitors applications.
- *CDN- maintains cache copies of data to speed up computations.*
- *Applications and data represent the primary workloads and resources hosted and managed in Azure, including web applications, databases, analytics solutions, and more.*
- *SQL Azure* - a cloud-based version of the SQL Server.

Azure



Open-source platforms for private clouds

- *Eucalyptus* - can be regarded as an open-source counterpart of Amazon's EC2.
- It provides an IaaS solution that allows organizations to create and manage their own cloud environments using their existing hardware.
- *Open-Nebula* - a private cloud with users actually logging into the head node to access cloud functions. The system is centralized and its default configuration uses the NFS file system.
- *Nimbus* - a cloud solution for scientific applications based on Globus software; inherits from Globus:
 - The image storage.
 - The credentials for user authentication.

Open-source platforms for private clouds

- *Eucalyptus* - can be regarded as an open-source counterpart of Amazon's EC2.
- It provides an IaaS solution that allows organizations to create and manage their own cloud environments using their existing hardware.

Example: A research institution deploys Eucalyptus in its data center to create a private cloud environment for running scientific simulations. Researchers can use Eucalyptus APIs to provision virtual machines, storage, and networking resources on-demand, similar to AWS EC2, for their computational experiments.

- *Open-Nebula* - a private cloud with users actually logging into the head node to access cloud functions. The system is centralized, and its default configuration uses the NFS file system.
- Example- A company uses OpenNebula to build a private cloud infrastructure for its internal IT operations. Employees log in to the central OpenNebula interface to request virtual machines, storage volumes, and network configurations for their development, testing, and production environments.

- *Nimbus* - a cloud solution for scientific applications based on Globus software; inherits from Globus:
 - The image storage.
 - The credentials for user authentication.
- A university research project utilizes Nimbus to deploy a cloud-based environment for processing large-scale genomics data. Nimbus integrates with Globus for secure data storage and user authentication, and it enables researchers to run bioinformatics algorithms on distributed compute nodes via SSH connections.

Eucalyptus : components

- *Virtual Machines* - run under several VMMs including Xen, KVM, and VMware.
- *Node Controller* - runs on server nodes hosting a VM and controls the activities of the node.
- *Cluster Controller* - controls a number of servers.
- *Cloud Controller* - provides the cloud access to end-users, developers, and administrators.
- *Storage Controller* - provides persistent virtual hard drives to applications. It is the correspondent of EBS.
- *Storage Service (Walrus)* - provides persistent storage; similar to S3, it allows users to store objects in buckets.

[EUCALYPTUS](#)
[RIGHTSCALE
MYCLOUD](#)
[ECOSYSTEM TOOLS](#)
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DOWNLOAD EUCALYPTUS

First time using Eucalyptus? Try [Eucalyptus FastStart](#).

1. Download and Install Eucalyptus

Choose a distribution:

[CentOS 5](#)
[CentOS 6](#)
[RHEL 5](#)
[RHEL 6](#)
[Ubuntu 10.04 LTS](#)
[Ubuntu 12.04 LTS](#)
[Source](#)
[Versions prior to Eucalyptus 3.1](#)
[Nightlies](#)
[Release Notes](#)

Looking for [Euca2ools](#)?

2. Configure Your Cloud

[Documentation](#)
[Engage \(Q&A\)](#)
[Consulting](#)
[Education](#)
[Support](#)

3. Use Your Cloud

To help get you started, we have prepared pre-packaged virtual machines ready to run in your Eucalyptus cloud.

[Download images](#)

Or check out a variety of [use cases](#).

Learn About Eucalyptus For

[MANAGERS](#)
[ARCHITECTS](#)
[APPLICATION ARCHITECTS](#)
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Euca2ools

Eucalyptus supported command-line tools.

[Get Euca2ools](#)

Ecosystem Tools

Find tools developed for Amazon EC2 and S3 which are compatible with Eucalyptus.

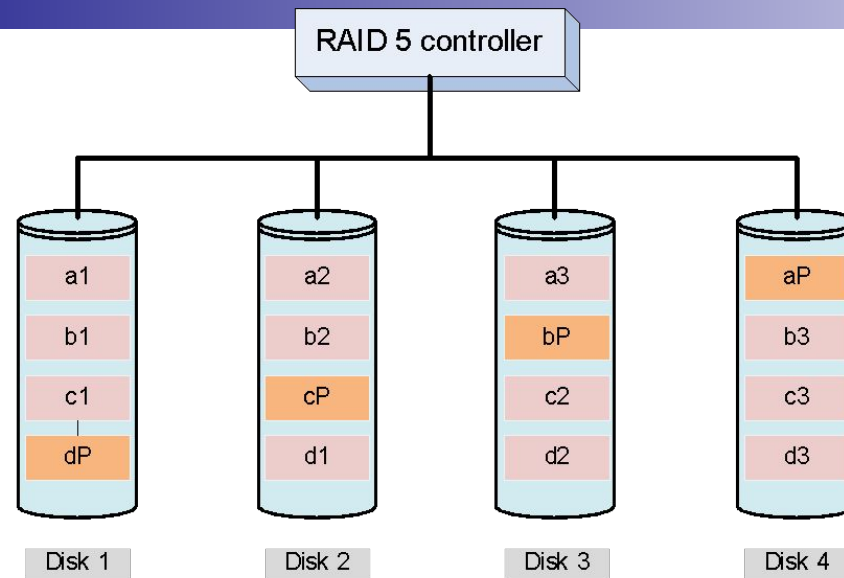
[Get tools](#)

Table 5: A side-by-side comparison of *Eucalyptus*, *OpenNebula*, and *Nimbus*.

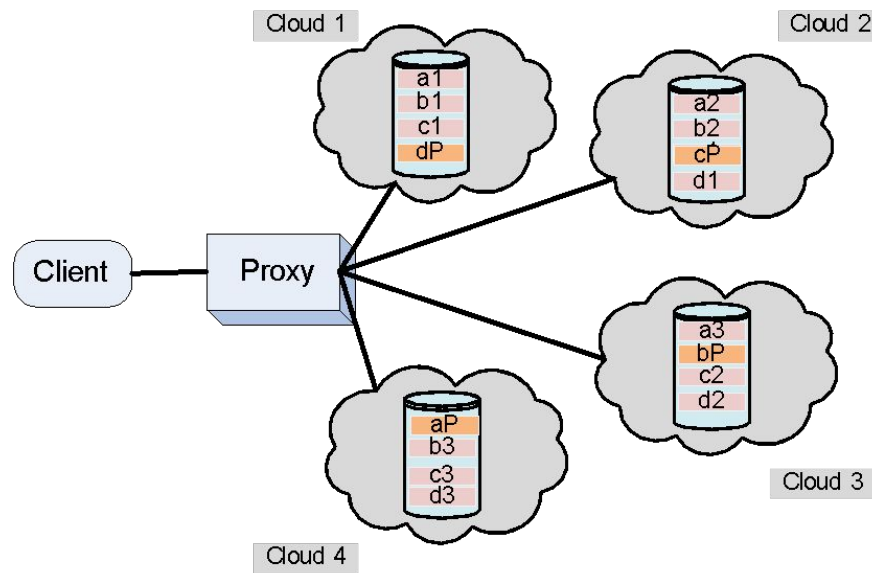
	<i>Eucalyptus</i>	<i>OpenNebula</i>	<i>Nimbus</i>
Design	Emulate EC2	Customizable	Based on Globus
Cloud type	Private	Private	Public/Private
User population	Large	Small	Large
Applications	All	All	Scientific
Customizability	Administrators limited users	Administrators and users	All but image storage and credentials
Internal security	Strict	Loose	Strict
User access	User credentials	User credentials	x509 credentials
Network access	To cluster controller	-	To each compute node

Cloud storage diversity and vendor lock-in

- Risks when a large organization relies on a single cloud service provider:
 - Cloud services may be unavailable for a short or an extended period of time.
 - Permanent data loss in case of a system failure.
 - The provider may increase the prices for service.
- Switching to another provider could be very costly due to the large volume of data to be transferred from the old to the new provider.
- A solution is to replicate the data to multiple cloud service providers, similar to data replication in RAID.



(a)



(b)

Cloud interoperability; the Intercloud

- cloud interoperability is the capacity or extent at which one cloud service is connected with the other by trading data as per strategy to get results
- An Intercloud □ a federation of clouds that cooperate to provide a better user experience.
- Cloud interoperability is the ability of applications and services developed on one platform to be used on another platform.
- Is an Intercloud feasible?
- Not likely at this time:
 - There are no standards for either storage or processing.
 - The clouds are based on different delivery models.
 - The set of services supported by these delivery models is large and open; new services are offered every few months.
 - Security is a major concern for cloud users and an Intercloud could only create new threats.

Energy use and ecological impact

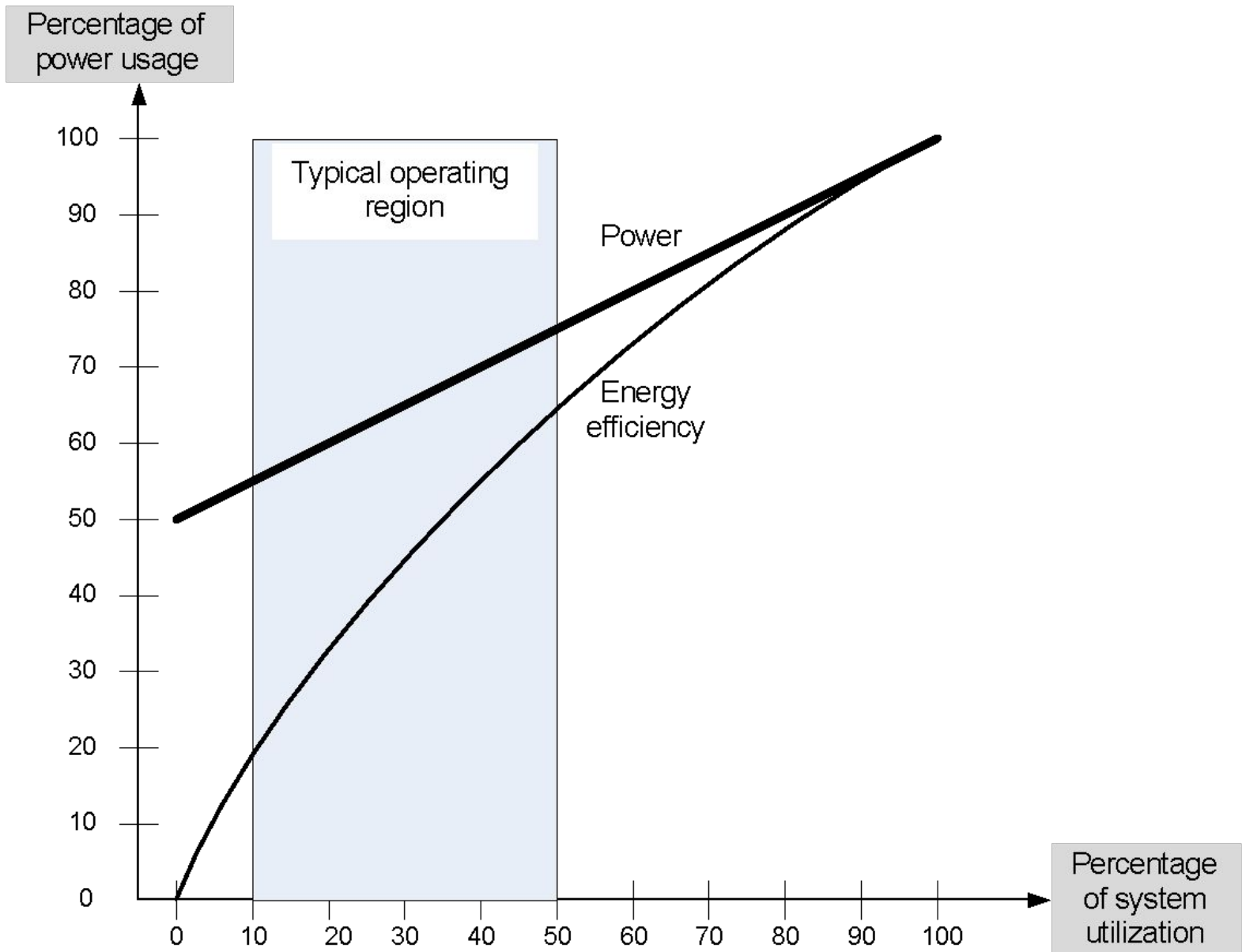
- The energy consumption of large-scale data centers and their costs for energy and for cooling are significant.
- In 2006, the 6,000 data centers in the U.S consumed 61×10^9 KWh of energy, 1.5% of all electricity consumption, at a cost of \$4.5 billion.
- The energy consumed by the data centers was expected to double from 2006 to 2011 and peak instantaneous demand to increase from 7 GW to 12 GW.
- The greenhouse gas emission due to the data centers is estimated to increase from 116×10^9 tones of CO_2 in 2007 to 257 tones in 2020 due to increased consumer demand.
- The effort to reduce energy use is focused on computing, networking, and storage activities of a data center.

Energy use and ecological impact (cont'd)

- Operating efficiency of a system is captured by the *performance per Watt of power*.
- The performance of supercomputers has increased 3.5 times faster than their operating efficiency – 7,000% versus 2,000% during the period 1998 – 2007.
- A typical Google cluster spends most of its time within the 10-50% CPU utilization range; there is a mismatch between server workload profile and server energy efficiency.

Energy-proportional systems

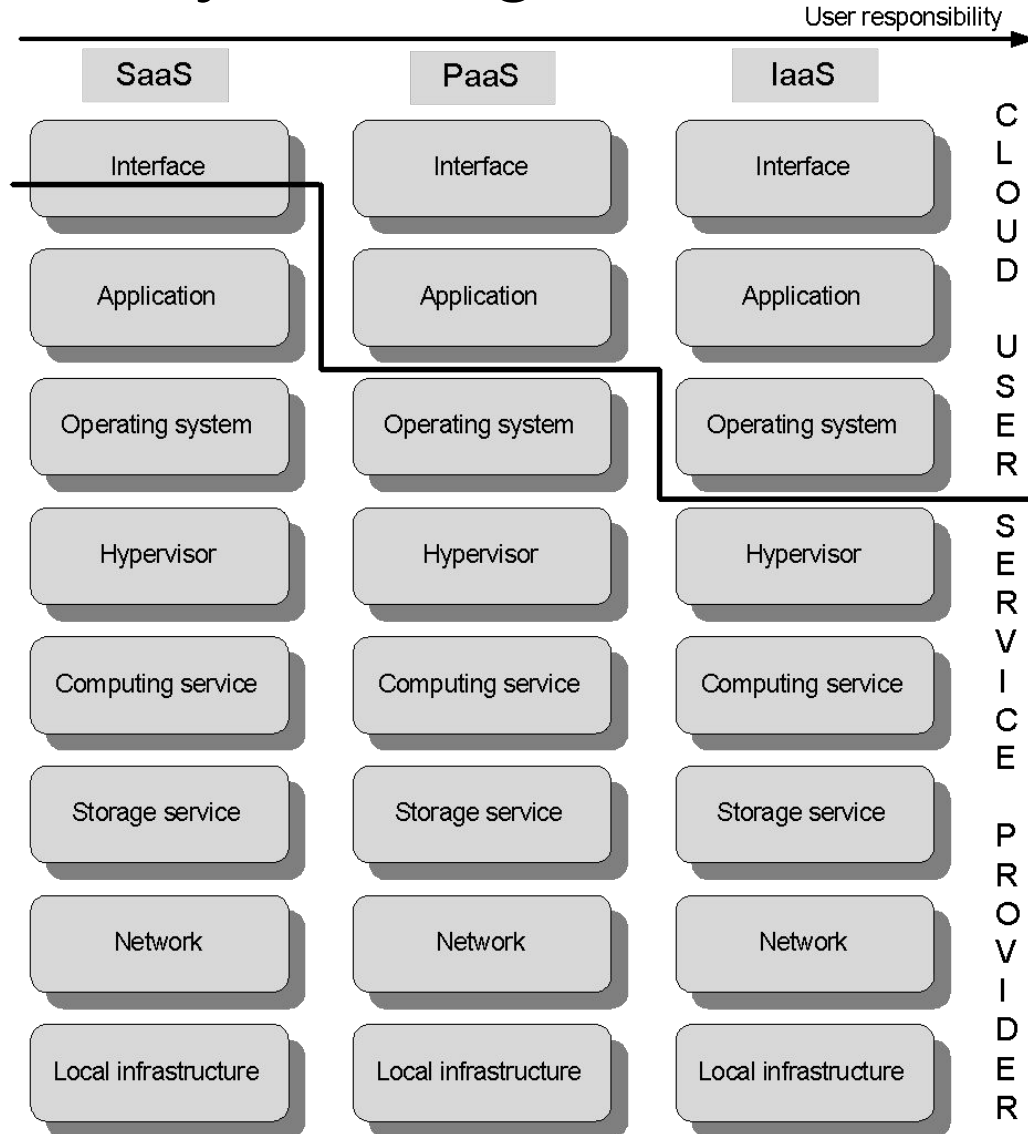
- An energy-proportional system consumes no power when idle, very little power under a light load and, gradually, more power as the load increases.
- By definition, an ideal energy-proportional system is always operating at 100% efficiency.
- Humans are a good approximation of an ideal energy proportional system; about 70 W at rest, 120 W on average on a daily basis, and can go as high as 1,000 – 2,000 W during a strenuous, short time effort.
- Even when power requirements scale linearly with the load, the energy efficiency of a computing system is not a linear function of the load; even when idle, a system may use 50% of the power corresponding to the full load.



Service Level Agreement (SLA)

- SLA - a negotiated contract between the customer and CSP; can be legally binding or informal. Objectives:
 - Identify and define the customer's needs and constraints including the level of resources, security, timing, and QoS.
 - Provide a framework for understanding; a critical aspect of this framework is a clear definition of classes of service and the costs.
 - Simplify complex issues; clarify the boundaries between the responsibilities of clients and CSP in case of failures.
 - Reduce areas of conflict.
 - Encourage dialog in the event of disputes.
 - Eliminate unrealistic expectations.
- Specifies the services that the customer receives, rather than how the cloud service provider delivers the services.

Responsibility sharing between user and CSP



User security concerns

- Potential loss of control/ownership of data.
- Data integration, privacy enforcement, data encryption.
- Data remanence after de-provisioning.
- Multi tenant data isolation.
- Data location requirements within national borders.
- Hypervisor security.
- Audit data integrity protection.
- Verification of subscriber policies through provider controls.
- Certification/Accreditation requirements for a given cloud service.

Reasons driving decision to use public clouds

Reason	Percentage who agree
Improved system reliability and availability	50%
Pay only for what you use	50%
Hardware savings	47%
Software license saving	46%
Lower labor costs	44%
Lower maintenance costs	42%
Reduced IT support needs	40%
Ability to take advantage of the latest functionality	40%
Less pressure on internal resources	39%
Solve problems related to updating/upgrading	39%
Rapid deployment	39%
Ability to scale up resources to meet the needs	39%
Ability to focus on core competencies	38%
Take advantage of the improved economics of scale	37%
Reduced infrastructure management needs	37%
Lower energy costs	29%
Reduced space requirements	26%
Create new revenue streams	23%